



MASTER'S THESIS

Construction of optimal portfolios in incomplete markets

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Abstract

In an incomplete market, no single pricing rule identifies an optimal investment strategy : the family of equivalent martingale measures is infinite, and the investor's own risk preferences become an unavoidable part of the answer. This thesis develops the convex duality theory that resolves this problem – discrete-time and multi-period (following Pliska), continuous-time and general (following Schachermayer and Kramkov-Schachermayer), and under convex portfolio constraints (following Cvitanić-Karatzas, connected to super-hedging via Bouchard-Cassagneux) – establishing, at the level of the duality theorem itself, that the unconstrained theory is the special case $K = R^d$ of the constrained one, and that the logarithmic investor's classical link to the growth-optimal portfolio and the minimal entropy martingale measure extends, in continuous time, to the exponential-utility case.

This theory is complemented by a fully tested numerical implementation (package `optiport`, 98 tests, five executed notebooks), validated, for portfolio constraints first and the general case second as required by the thesis guidelines, through four experiments : exact pathwise coincidence between Merton's dynamic-programming solution and the continuous-time duality first-order condition in the Black-Scholes model ; a discrete-time distinction, made numerically visible, between Samuelson's myopic-policy theorem and Merton's intertemporal hedging demand ; identification of the minimal entropy martingale measure in an incomplete jump-diffusion market ; and pathwise validation of the constrained duality theorem across every qualitative regime of a box constraint, cross-checked against an independently coded numerical optimiser. Two bugs found during development – both traceable to a missing additive term in the exponent of a state-price density – are documented as regression tests and discussed as evidence for the value of closed-form, pathwise validation wherever it is available.

Keywords : incomplete markets, expected utility maximisation, convex duality, martingale measures, portfolio constraints, minimal entropy, Kramkov-Schachermayer, backward stochastic differential equations.

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Chapitre 1

Introduction

The construction of an optimal investment strategy is one of the oldest and most central questions in mathematical finance. Since the seminal work of Markowitz on mean-variance portfolio selection and the dynamic extension proposed by Merton, the problem has been formulated as the maximisation of an investor's expected utility of terminal wealth, subject to the constraint that the portfolio is financed by an initial capital and traded without withdrawing or injecting further funds. This formulation is elegant and, in the particular setting of complete markets, admits an essentially explicit solution : the Black-Scholes model and its discrete-time counterpart, the binomial model, both possess a unique equivalent martingale measure, and every contingent claim can be perfectly replicated by a self-financing trading strategy. In such a setting, the optimal portfolio problem reduces, through a simple change of measure, to a static problem of allocating a fixed budget across a set of Arrow-Debreu securities.

This elegance, however, rests on an assumption that real markets systematically violate : completeness. As soon as the source of randomness driving asset prices is richer than the number of tradable instruments available to hedge it – because of jumps, stochastic volatility, transaction costs, or simply because certain risks are not attached to any traded security – the market becomes incomplete. Perfect replication of an arbitrary claim is then generally impossible, and, more importantly for the purpose of this work, the martingale measure ceases to be unique. Instead of a single pricing rule, the market admits an entire family of equivalent martingale measures, each consistent with the absence of arbitrage. An investor can no longer simply "read off" the optimal portfolio from a unique risk-neutral measure : she must genuinely solve an optimisation problem, and the choice of criterion – her utility function – becomes an unavoidable part of the answer.

A further layer of realism, and one that is central to this thesis, comes from the constraints actually faced by an investor. Trading restrictions such as the prohibition of short sales, bounds on leverage, or the presence of non-tradable endowments are the rule rather than the exception in practice. These constraints turn the set of admissible trading strategies from a vector space into a closed convex set, which in turn changes the mathematical nature of the optimisation problem. As will be discussed at length in this work, the unconstrained problem is, from a purely mathematical standpoint, best understood as a special case of the constrained one, rather than the other way around.

The central object of study is therefore the maximisation, over a suitable class of admissible self-financing strategies, of the expected utility of terminal wealth generated by a given initial capital. Solving this problem directly is, in general, difficult : the set of terminal wealths that can be reached from a given budget is not easy to describe explicitly once the market is incomplete or constrained. The approach that has proved most fruitful over the last three decades, and which forms the methodological backbone of this thesis, is the use of convex duality. Rather than attacking the primal problem head-on, one associates to it a dual problem posed over the family of equivalent martingale measures (or, in the constrained case, over a suitably enlarged family of measures penalised by the support function of the constraint set). The link between the two problems is made precise by results going back to the fundamental theorem of asset pricing, and by the convex duality theory developed for expected utility maximisation in incomplete markets, notably by Karatzas, Lehoczky, Shreve and Xu, by Cvitanić and Karatzas for the constrained case, and by Kramkov and Schachermayer in a general setting.

This thesis aims to give a self-contained and rigorous account of this theory, and to complement it with concrete numerical methods for solving the optimal portfolio problem in practice. Three reference sources structure the presentation. Pliska's textbook provides the discrete-time, multi-period framework, in which the duality argument can be carried out by elementary backward induction and made fully explicit. Schachermayer's lecture notes provide the continuous-time generalisation, together with the precise conditions – notably an asymptotic elasticity condition on the utility function – under which the duality theorem holds. Bouchard and Cassagneux's textbook connects this theory of optimal investment to the closely related, but conceptually distinct, problem of super-hedging under constraints, and provides the bridge towards backward stochastic equations, which will be used as one of the numerical tools of this work.

The remainder of this thesis first reviews the relevant literature, from Merton's original dynamic programming approach to the martingale and duality methods that followed. It then develops the discrete-time, multi-period theory following Pliska, establishing the primal-dual correspondence by dynamic programming, before turning to the continuous-time setting following Schachermayer and to its connection with the Hamilton-Jacobi-Bellman approach to stochastic control. The theory is subsequently extended to the case of convex portfolio constraints, following Cvitanić and Karatzas, together with a discussion of its relation to super-hedging. Optimisation algorithms and numerical schemes are then developed, first for the constrained case and then for the unconstrained case, as required by the thesis guidelines, and are illustrated through a series of numerical experiments. The thesis closes with a discussion of the limitations of the framework and of possible extensions.

Chapitre 2

Litterature Review

The problem of optimal portfolio choice under uncertainty has been studied from several, progressively more general, angles over the past half-century. This chapter traces the main lines of this development, from the earliest dynamic formulations to the general convex duality theory that underlies the approach adopted in this thesis, and situates the three reference works – Pliska (1), Schachermayer (2), and Bouchard and Cassagneux (3) – within this broader picture.

2.1 From static allocation to dynamic control

The modern theory of portfolio choice begins with Markowitz’s mean-variance framework (4), in which an investor allocates wealth across risky assets so as to trade off expected return against variance, at a single date. Merton (5; 6) extended this idea to a genuinely dynamic setting, formulating the investor’s problem as one of stochastic control : wealth is invested continuously over time, and the investor seeks to maximise the expected utility of terminal wealth (or of consumption along the way). Working in a complete market driven by Brownian motion, Merton solved this problem through the Hamilton-Jacobi-Bellman equation and obtained closed-form optimal strategies for power and logarithmic utility. This dynamic programming approach remains, to this day, the natural starting point for continuous-time portfolio problems, and its connection to the duality approach developed later in this thesis is made explicit when the two are shown to agree in the classical Black-Scholes setting.

2.2 Completeness and the martingale approach

In parallel, the fundamental theorem of asset pricing, established in its discrete-time form by Harrison and Kreps (7) and given its definitive treatment for multi-period markets by Harrison and Pliska (8), clarified the link between the absence of arbitrage and the existence of an equivalent martingale measure. When the market is complete, this measure is unique, every contingent claim can be replicated, and the portfolio problem reduces to a static allocation problem under the risk-neutral measure. This martingale approach to portfolio choice was carried into continuous-time by Cox and Huang (9) and by Karatzas, Lehoczky and Shreve (10), who reformulated Merton's dynamic control problem as a static optimisation over terminal wealth, subject to a budget constraint expressed under the martingale measure. In a complete market, this static reformulation is exactly equivalent to Merton's dynamic programming solution, and provides a first, elementary instance of the primal-dual correspondence that this thesis develops in full generality.

2.3 Incompleteness and the martingale-duality method

The martingale approach becomes substantially more delicate once the market is incomplete, because the equivalent martingale measure is no longer unique : the investor is no longer choosing among a fixed set of state prices, but implicitly also among the members of an entire family of admissible pricing measures. Karatzas, Lehoczky, Shreve and Xu (11) addressed this difficulty by formulating a dual problem over the set of equivalent martingale measures, and showed that the optimal terminal wealth of the primal problem could be recovered from the solution of this dual problem together with a suitably chosen dual optimiser. This martingale-duality method is the direct ancestor of the approach followed in this thesis, and it is the one presented, in its multi-period and discrete-time form, in Pliska's (1) treatment of the subject, where the argument can be carried out by elementary backward induction and made fully explicit at every step.

2.4 General convex duality

The martingale-duality method of Karatzas, Lehoczky, Shreve and Xu was formulated under fairly restrictive assumptions on the utility function and on the market model. Kramkov and Schachermayer (12) placed the theory on a substantially more general

footing, working with an abstract convex duality between the utility function and its Legendre-Fenchel conjugate, and identifying a precise condition on the utility function – an asymptotic elasticity condition – under which existence and uniqueness of both the primal and the dual optimisers can be guaranteed, together with the absence of a duality gap. Schachermayer’s lecture notes (2), one of the three reference works of this thesis, give a systematic and self-contained account of this general theory, and are followed closely in the continuous-time part of this work.

2.5 Portfolio constraints

A separate strand of the literature addresses the case where the investor is not free to trade arbitrarily, but is instead restricted to a closed convex set of admissible positions – for instance because short sales are prohibited, because leverage is bounded, or because part of the investor’s endowment cannot be traded. Cvitanić and Karatzas (13) showed that this constrained problem admits a dual formulation as well, but one posed over an enlarged family of auxiliary measures, obtained by penalising the market price of risk with the support function of the constraint set. Unconstrained markets then appear as the particular case in which the constraint set is the whole space and the support function vanishes identically. This constrained duality is closely related to the theory of superhedging under constraints, developed among others by El Karoui and Quenez (14) and, in the form used in this thesis, by Bouchard and Cassagneux (3), who connect it to the theory of backward stochastic differential equations. This connection provides, as will be seen later, a natural route towards the numerical solution of the constrained portfolio problem.

2.6 Positioning of this thesis

Table 2.1 summarises the main approaches reviewed above along four dimensions : the assumptions placed on the utility function, whether the underlying market is required to be complete, whether portfolio constraints are accommodated, and the type of existence result obtained.

The picture that emerges is one of increasing generality along two largely independent axes : from complete to incomplete markets, and from unconstrained to constrained trading. This thesis follows the natural path suggested by this table. The discrete-time, multi-

Approach	Utility	Market	Constraints	Existence
Merton (dynamic programming) (5; 6)	Power, log	Complete	No	Explicit solution
Cox-Huang, Karatzas-Lehoczky-Shreve (static martingale) (9; 10)	Power, log, general concave	Complete	No	Explicit via change of measure
Karatzas-Lehoczky-Shreve-Xu (martingale duality) (11)	General concave	Incomplete	No	Under integrability conditions
Cvitanic-Karatzas (constrained duality) (13)	General concave	Incomplete	Yes (convex)	Under integrability conditions
Kramkov-Schachermayer (general duality) (12)	Asymptotic elasticity < 1	Incomplete	No	Full existence-uniqueness

TABLE 2.1 – Comparison of the main approaches to expected utility maximisation.

period case is treated first, following Pliska (1), because the duality argument there can be made completely explicit through backward induction, without requiring the more delicate integrability and elasticity conditions of the continuous-time theory. The continuous-time case is then developed following Schachermayer (2), establishing the precise conditions under which the general duality theorem holds and connecting it to the Hamilton-Jacobi-Bellman approach of Merton (5). Portfolio constraints are then introduced following Cvitanic and Karatzas (13), together with their connection, via Bouchard and Cassagneux (3), to super-hedging and backward stochastic equations – a connection that is exploited directly in the numerical part of this work, where constrained algorithms are developed before their unconstrained counterparts are recovered as a special case.

Chapitre 3

Discrete Times

This chapter develops the discrete-time, multi-period theory of optimal portfolio choice in incomplete markets, following Pliska (1). Unlike the introductory chapter, the presentation here is fully formal : the discrete-time setting is precisely where the duality argument can be carried out by elementary backward induction, with every step made explicit, and this rigour is what makes it the natural starting point of the thesis. (Note : this chapter assumes full mathematical formalism is appropriate here, in contrast with the textual introduction – flag if a lighter style is preferred instead.)

3.1 Market model and admissible strategies

Fix a finite time horizon $T \in \mathbb{N}$ and a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{t=0}^T, P)$, with $\mathcal{F}_0$ trivial and Ω finite, so that every random variable is bounded and every conditional expectation is a finite sum. The market consists of one riskless asset, taken as numeraire, and d risky assets whose discounted price process is $S = (S_t)_{t=0}^T = (S_t^1, \dots, S_t^d)_{t=0}^T$, adapted to $(\mathcal{F}_t)$.

A *trading strategy* is a predictable process $\theta = (\theta_t)_{t=1}^T$, $\theta_t \in \mathbb{R}^d$, where θ_t is the number of shares of each risky asset held over the period $(t-1, t]$. Given an initial capital $x \geq 0$, the self-financing wealth process associated with θ is

$$X_t^{x, \theta} = x + \sum_{s=1}^t \theta_s \cdot (S_s - S_{s-1}), \quad t = 0, \dots, T. \quad (3.1)$$

A strategy θ is *admissible* for the initial capital x , written $\theta \in \mathcal{A}(x)$, if $X_t^{x, \theta} \geq 0$ for every t . This lower bound rules out unbounded borrowing against future gains and is the discrete-time analogue of the admissibility condition used later in continuous time.

3.2 Absence of arbitrage and the fundamental theorem of asset pricing

The market admits an *arbitrage opportunity* if there exists $\theta \in \mathcal{A}(0)$ such that $X_T^{0,\theta} \geq 0$ almost surely and $P(X_T^{0,\theta} > 0) > 0$.

Let $\mathcal{M}_e$ denote the set of probability measures Q equivalent to P under which S is a martingale with respect to $(\mathcal{F}_t)$.

[Discrete-time fundamental theorem of asset pricing] The market is arbitrage-free if and only if $\mathcal{M}_e \neq \emptyset$.

This is the discrete, finite- Ω version of the theorem established by Harrison and Pliska (8); the proof, which reduces to a separation argument between the cone of attainable claims and the positive orthant at each date, is given in full in Pliska's textbook (1) and is not repeated here. What matters for the sequel is the following dichotomy.

The market is *complete* if every $\mathcal{F}_T$ -measurable random variable is attainable, i.e. can be written as $x + \sum_{t=1}^T \theta_t \cdot (S_t - S_{t-1})$ for some x and some predictable θ . The market is *incomplete* otherwise.

An arbitrage-free market is complete if and only if $\mathcal{M}_e$ is a singleton. It is precisely the case $|\mathcal{M}_e| > 1$ that is of interest in this thesis : the investor's optimal strategy can no longer be read off from a single change of measure, and a genuine optimisation over $\mathcal{M}_e$ becomes part of the problem.

3.3 The multi-period utility maximisation problem

Let $U : (0, \infty) \rightarrow \mathbb{R}$ be a strictly increasing, strictly concave, continuously differentiable utility function satisfying the Inada conditions $U'(0^+) = \infty$ and $U'(\infty) = 0$. For an initial capital $x > 0$, the *primal problem* is

$$u(x) = \sup_{\theta \in \mathcal{A}(x)} E[U(X_T^{x,\theta})]. \quad (3.2)$$

Because $\mathcal{A}(x)$ is a convex set and U is concave, the primal problem is a concave maximisation problem; its difficulty lies not in convexity but in the fact that the set of attainable terminal wealths, $\{X_T^{x,\theta} : \theta \in \mathcal{A}(x)\}$, does not admit a simple description once the market is incomplete.

3.4 Convex duality

Let $\tilde{V}$ denote the convex conjugate of $-U(-\cdot)$, equivalently

$$\tilde{V}(y) = \sup_{x>0} (U(x) - xy), \quad y > 0, \quad (3.3)$$

a decreasing, convex function satisfying $\tilde{V}'(y) = -I(y)$, where $I = (U')^{-1}$. For $y > 0$ define the *dual value function*

$$\hat{v}(y) = \inf_{Q \in \mathcal{M}_e} E \left[\tilde{V} \left(y \frac{dQ}{dP} \right) \right]. \quad (3.4)$$

The following theorem is the discrete-time counterpart of the general duality results that will be developed in continuous time.

[Primal-dual duality, finite horizon] For every $x > 0$,

$$u(x) = \inf_{y>0} (\hat{v}(y) + xy), \quad (3.5)$$

and the infimum is attained at some $y^* = y^*(x) > 0$ and some $Q^* \in \mathcal{M}_e$ attaining the infimum in (3.4). Moreover, the optimal terminal wealth in the primal problem (3.2) is

$$X_T^{x,\theta^*} = I \left(y^* \frac{dQ^*}{dP} \right), \quad (3.6)$$

and the pair (y^*, Q^*) satisfies the budget-binding condition $E^{Q^*}[X_T^{x,\theta^*}] = x$.

Equation (3.6) is the discrete-time first-order condition linking the optimal terminal wealth to the optimal dual measure Q^* : the marginal utility of optimal terminal wealth is proportional, state by state, to the density of the optimal martingale measure with respect to P . This is the precise sense in which Q^* – and not an arbitrary element of $\mathcal{M}_e$ – is the "correct" pricing measure for the investor's own problem.

3.4.1 Proof by backward induction

The finite-horizon duality theorem can be proved constructively by dynamic programming, which has the merit of also producing an implementable algorithm. Define the value

functions $J_t : R_+ \rightarrow R$, $t = T, T-1, \dots, 0$, by backward induction :

$$J_T(x) = U(x), \quad (3.7)$$

$$J_t(x) = \sup_{\theta_t \in R^d} E[J_{t+1}(x + \theta_t \cdot (S_t - S_{t-1})) \mid \mathcal{F}_{t-1}], \quad t = T, \dots, 1. \quad (3.8)$$

Standard arguments (measurable selection, using the finiteness of Ω) show that $J_0(x) = u(x)$ and that an optimal strategy θ^* is recovered from the arguments attaining the successive suprema. Each J_t inherits concavity from U , and its convex conjugate $\tilde{J}_t$ satisfies its own backward recursion ; comparing this recursion, date by date, with the one satisfied by $E[\tilde{V}(y dQ/dP) \mid \mathcal{F}_{t-1}]$ under the martingale constraint on Q identifies $\hat{v}$ as the initial value of this dual recursion, which yields the theorem. The details of this comparison, worked out explicitly on a two- and three-period example, are given in the numerical chapter of this thesis, together with the corresponding implementation.

3.5 The logarithmic case : entropy and growth-optimal portfolios

The case $U(x) = \log x$ deserves special attention, both because it is analytically the most tractable and because of its connection to the entropy-based methods used elsewhere in this line of work. For $U = \log$, the conjugate is $\tilde{V}(y) = -\log y - 1$, and the dual problem (3.4) becomes, up to the additive constant and a change of sign,

$$\inf_{Q \in \mathcal{M}_e} E \left[\log \frac{dQ}{dP} \right] = \inf_{Q \in \mathcal{M}_e} D_{\text{KL}}(Q \parallel P), \quad (3.9)$$

the relative entropy of Q with respect to P . The optimal dual measure Q^* is therefore the *minimal entropy martingale measure* in $\mathcal{M}_e$, and by (3.6) the optimal terminal wealth satisfies $X_T^{x, \theta^*} \propto (dQ^*/dP)^{-1}$: the logarithmic investor's optimal strategy is exactly the growth-optimal (Kelly) portfolio, and its associated pricing measure is obtained by minimising relative entropy over $\mathcal{M}_e$ rather than by postulating it a priori. This is the same variational problem that appears, from the opposite direction, in entropy-regularised calibration under a fixed pricing constraint ; here the roles of prior and constraint are reversed, but the underlying Legendre duality is identical.

Chapitre 4

Continuous Time

This chapter extends the multi-period theory developed in the previous chapter to a continuous-time, general semimartingale setting, following Schachermayer's lecture notes (2), which are themselves built on the duality theory of Kramkov and Schachermayer (12). The same formalism as the previous chapter is used throughout. (As before : full mathematical formalism, with statements and structured proof sketches ; flag if a lighter style is preferred.)

4.1 Market model and admissibility

Let $S = (S_t)_{0 \leq t \leq T}$ be a locally bounded, R^d -valued semimartingale on a filtered probability space $(\Omega, \mathcal{F}, (\mathcal{F}_t)_{0 \leq t \leq T}, P)$, representing the discounted prices of d risky assets, the riskless asset being taken as numéraire. A self-financing portfolio is a pair (x, H) , where H is a predictable, S -integrable process, with associated wealth process $X_t = x + (H \cdot S)_t = x + \int_0^t H_u dS_u$. As in the discrete case, some lower bound on wealth is needed to exclude doubling strategies : H is *admissible* if there is a constant $C \geq 0$ such that $(H \cdot S)_t \geq -C$ almost surely, for all t .

Let $\mathcal{M}_e(S)$ denote the set of probability measures $Q \sim P$ under which S is a (local) martingale. The no-arbitrage condition used throughout – no free lunch with vanishing risk, in the sense of Delbaen and Schachermayer (15) – is equivalent, by the general fundamental theorem of asset pricing, to $\mathcal{M}_e(S) \neq \emptyset$: this is the continuous-time analogue of the discrete result of the previous chapter, and it is assumed to hold throughout.

4.2 Two classes of utility functions

Following Schachermayer (2), two cases are distinguished according to the domain of the utility function U .

- **Case 1** (negative wealth excluded) : $\text{dom}(U) = (0, \infty)$, U strictly increasing, strictly concave, continuously differentiable, with $U'(0^+) = \infty$ and $U'(\infty) = 0$. The power utilities $U(x) = x^\alpha/\alpha$, $\alpha \in (-\infty, 1) \setminus \{0\}$, and $U(x) = \log x$ belong to this class.
- **Case 2** (negative wealth allowed) : $\text{dom}(U) = \mathbb{R}$, with $U'(-\infty) = \infty$ and $U'(\infty) = 0$. The exponential utility $U(x) = -\gamma^{-1}e^{-\gamma x}$, $\gamma > 0$, is the standard example.

The two cases require different technical treatments of the dual problem, as will be seen below, but the qualitative conclusions – existence, uniqueness, and a first-order duality relation between the primal and dual optimizers – are parallel.

4.3 The primal problem and the enlarged dual domain

For $x > 0$ (Case 1) or $x \in \mathbb{R}$ (Case 2), the primal value function is

$$u(x) = \sup_{H \text{ admissible}} E[U(x + (H \cdot S)_T)]. \quad (4.1)$$

A first difficulty, absent in the finite-dimensional discrete case, is that the set of terminal wealths dominated by some admissible strategy,

$$\mathcal{C}(x) = \{X_T \in L_+^0 : X_T \leq x + (H \cdot S)_T \text{ for some admissible } H\} \quad (4.2)$$

(Case 1), is naturally a subset of $L_+^0(\Omega, \mathcal{F}_T, P)$, the space of nonnegative random variables with the topology of convergence in probability – a space with no useful duality with $L^1(P)$ in general. The resolution, due to Kramkov and Schachermayer (12), is to work with the pairing $\langle f, g \rangle = E[fg] \in [0, \infty]$, well defined (possibly infinite) for $f, g \geq 0$, and to enlarge the natural dual domain $\mathcal{M}_a(S)$ (absolutely continuous local martingale measures) to its closed, convex, solid hull $\mathcal{D} \subset L_+^0(P)$. Elements of $\mathcal{D}$ can be described as random variables dominated by an almost-sure limit of a sequence of densities dQ_n/dP , $Q_n \in \mathcal{M}_a(S)$; in particular they need not integrate to one. This enlargement is not a technical nicety : the dual optimizer for the enlarged problem may genuinely fail to be the density of a probability measure even for the logarithmic utility on a perfectly regular diffusion model,

which is why $\mathcal{D}$, rather than $\mathcal{M}_a(S)$ itself, is the correct domain of the dual problem in Case 1.

In Case 2 the situation is comparatively simpler : because U is finite on all of R , the appropriate enlargement of the primal domain is instead obtained by an L^1 -closure construction with respect to U itself, but the dual optimizer, as shown by Bellini and Frittelli (16), remains a genuine element of $\mathcal{M}_a(S)$.

4.4 The reasonable asymptotic elasticity condition

Beyond the no-arbitrage assumption on S , existence of the dual optimizer and absence of a duality gap require a mild growth condition on U alone, introduced by Kramkov and Schachermayer (12).

[Asymptotic elasticity] The asymptotic elasticity of U at $+\infty$ is

$$AE_{+\infty}(U) = \limsup_{x \rightarrow \infty} \frac{xU'(x)}{U(x)}. \quad (4.3)$$

U has *reasonable asymptotic elasticity* if $AE_{+\infty}(U) < 1$ (and, in Case 2, additionally $\liminf_{x \rightarrow -\infty} xU'(x)/U(x) > 1$).

For any concave, increasing U one always has $AE_{+\infty}(U) \leq 1$; the substantive requirement is the strict inequality. Power utilities satisfy $AE_{+\infty}(U) = \alpha < 1$ and log satisfies $AE_{+\infty}(U) = 0$, so both are always admissible. The condition fails only for utility functions whose marginal utility decays to zero so slowly, relative to the average utility $U(x)/x$, that the two become asymptotically indistinguishable; Kramkov and Schachermayer (12) construct an explicit such U , differing arbitrarily little from an ordinary power utility for practical purposes, for which the dual optimizer fails to exist even in a market as simple as a complete diffusion model driven by Brownian motion. This condition is therefore a genuine restriction on U , not an artefact of an incomplete or otherwise pathological market model, and it will resurface as the binding technical assumption whenever the theory is applied later in this thesis.

4.5 The duality theorem

[Kramkov–Schachermayer duality (12)] Let S satisfy the no-arbitrage condition and let U satisfy the regularity conditions of Case 1 (respectively Case 2) together with reasonable

asymptotic elasticity. Define

$$u(x) = \sup_{X_T \in \mathcal{C}(x)} E[U(X_T)], \quad v(y) = \inf_{Y_T \in \mathcal{D}(y)} E[\tilde{V}(Y_T)] \quad (4.4)$$

(with $\mathcal{D}$ replaced by $\mathcal{M}_a(S)$ in Case 2). Then :

1. u and v are finite, continuously differentiable, strictly concave and convex respectively, and conjugate to one another ;
2. the optimizers $\hat{X}_T(x)$ and $\hat{Y}_T(y)$ exist, are unique, and satisfy the first-order condition

$$\hat{X}_T(x) = I(\hat{Y}_T(y)), \quad \hat{Y}_T(y) = U'(\hat{X}_T(x)), \quad (4.5)$$

where x and y are related by $u'(x) = y$;

3. in Case 2, if the dual optimizer is (the density of) an equivalent martingale measure $\hat{Q}(y)$, then $\hat{X}_T(x)$ is the terminal value of a genuine admissible trading strategy, and the process $\hat{X}_t(x)\hat{Y}_t(y)$ is a uniformly integrable martingale under P .

This is the exact continuous-time, general-semimartingale counterpart of the finite- Ω result established by dynamic programming in the previous chapter : the same first-order relation $U'(\hat{X}_T) = y d\hat{Q}/dP$ links the optimal terminal wealth to the optimal dual measure, only the functional-analytic machinery required to establish existence has changed, from a finite-dimensional compactness argument to the enlargement of the dual domain and a convex-analytic minimax argument in L_+^0 .

4.6 Connection with the Hamilton-Jacobi-Bellman approach

In the classical Black-Scholes model, which is complete, the set $\mathcal{M}_e(S)$ reduces to a singleton and the enlargement to $\mathcal{D}$ plays no role. In this special case, the martingale duality approach above and Merton's (5) original dynamic programming approach must agree, and this agreement provides a natural consistency check for the theory. Merton's approach values the portfolio problem through the value function $J(t, x) = \sup E[U(X_T) \mid X_t = x]$, which solves the Hamilton-Jacobi-Bellman equation obtained by dynamic programming, and the optimal proportion of wealth invested in the risky asset for CRRA utility $U(x) = x^\alpha/\alpha$ is then given by Merton's classical constant-mix formula, expressed in terms of the market price of risk and the risk-aversion parameter $1 - \alpha$. On the other hand, applying the duality theorem above with the unique martingale measure Q gives,

by (3.6)-type reasoning transposed to continuous time, $\hat{X}_T(x) = I(y dQ/dP)$, which, once dQ/dP is written explicitly in terms of the Black-Scholes market price of risk, reproduces exactly the same terminal wealth as Merton's verification argument. Recovering this coincidence explicitly, by computing both sides in the Black-Scholes model, is carried out in the numerical chapter of this thesis as the primary validation test of the implementation.

4.7 The exponential case and minimal entropy

As in the discrete-time chapter, the case of exponential utility, $U(x) = -\gamma^{-1}e^{-\gamma x}$ (Case 2), admits a particularly transparent dual problem : the conjugate is $\tilde{V}(y) = \gamma^{-1}y(\log y - 1)$, and the dual problem becomes, up to affine reparametrisation,

$$\inf_{Q \in \mathcal{M}_e(S)} D_{\text{KL}}(Q \parallel P), \quad (4.6)$$

so that the dual optimizer is again the *minimal entropy martingale measure*, exactly as the minimal entropy measure emerged from the logarithmic utility in discrete time, except that here the correspondence is with exponential rather than logarithmic utility – a distinction that is a genuine feature of the continuous-time theory and not a discrepancy with the previous chapter, since the two utilities play structurally different roles (numeraire/growth-optimal portfolio for log, versus indifference pricing for the exponential case). Existence of this minimal entropy measure, and equivalence of the optimizer to P , is guaranteed as soon as some $Q \in \mathcal{M}_e(S)$ has finite relative entropy with respect to P , by a classical result of Csiszár (17) transposed to this setting by Bellini and Frittelli (16), and by Frittelli (18) independently.

The results of this chapter, together with those of the previous one, give a complete primal-dual picture of unconstrained utility maximisation, in both discrete and continuous time. The next chapter extends this picture to the case where the investor is restricted to a closed convex set of admissible positions.

Chapitre 5

Constraint Optimization

This chapter extends the duality theory of the two preceding chapters to the case where the investor's trading strategies are restricted to a closed convex set, following Cvitanić and Karatzas (13), and discusses the connection of this constrained problem to super-hedging and to backward stochastic differential equations, following Bouchard and Cassagneux (3). As announced in the introduction, this is, mathematically, the general case of the theory : the unconstrained results of the two previous chapters are recovered as the particular instance in which the constraint set is the whole space. (Same formalism level as the previous two chapters.)

5.1 Convex constraints on trading strategies

Work in the continuous-time semimartingale setting of the previous chapter. Let $K \subset R^d$ be a closed convex set containing the origin, representing the constraint on the proportions of wealth $\pi_t \in R^d$ invested in the risky assets at each date (so that $\pi_t \in K$ for a.e. t , in place of the unconstrained $\pi_t \in R^d$ of the previous chapters). The set of K -admissible strategies is denoted $\mathcal{A}_K(x)$. Three examples recur throughout this chapter :

- $K = R_+^d$: no short sales of the risky assets are allowed ;
- $K = \{\pi : \mathbf{1}^\top \pi \leq \kappa\}$ for some $\kappa > 0$: leverage (total risky exposure) is bounded ;
- $K = R^{d_1} \times \{0\}^{d_2}$: a subset of d_2 assets is non-tradable, or an endowment in these assets cannot be hedged directly.

Unconstrained trading, treated in the previous chapters, corresponds to $K = R^d$.

5.2 The support function and the auxiliary market approach

The key device introduced by Cvitanić and Karatzas (13) is to associate to K its *support function*

$$\delta(\nu) = \sup_{\pi \in K} (-\pi \cdot \nu), \quad \nu \in R^d, \quad (5.1)$$

a nonnegative, convex, lower semicontinuous, positively homogeneous function, equal to 0 on the *barrier cone* (the polar cone of K) and to $+\infty$ outside it. For $K = R_+^d$, $\delta(\nu) = \max_i (-\nu_i)^+ = 0$ if $\nu \geq 0$ and $+\infty$ otherwise; for $K = R^d$ (no constraint), $\delta \equiv 0$ on all of R^d .

For each ν in the domain of δ (an admissible dual process, suitably integrable), define the *auxiliary market* obtained by shifting the interest rate and the drift of the risky assets by ν and $\delta(\nu)$ respectively. Concretely, if the original market has market price of risk λ_t , the auxiliary market associated with ν_t has market price of risk $\lambda_t + \sigma_t^{-1} \nu_t$ and an additional running cost $\delta(\nu_t)$ subtracted from wealth. Two observations drive the whole construction :

1. Because $\delta(\nu) \geq -\pi \cdot \nu$ for every $\pi \in K$, a K -constrained strategy's shifted wealth process is always a supermartingale under the martingale measure of the auxiliary market, for *every* admissible ν ;
2. Equality (i.e. a genuine martingale) holds along the constrained strategy that is optimal for the particular auxiliary market whose ν is, in a suitable sense, tangent to K at the optimal π_t .

This yields a representation of the constrained value function as an infimum, over auxiliary markets, of unconstrained value functions – exactly the shape of a minimax duality, but now with an extra layer compared to the unconstrained case : optimizing over ν plays the role that optimizing over $Q \in \mathcal{M}_e(S)$ played before, except that the auxiliary "measures" here are not necessarily martingale measures for the original S , but for the ν -shifted process.

5.3 The constrained duality theorem

[Cvitanić–Karatzas duality (13)] Let $\mathcal{D}_K$ denote the (enlarged, as in the unconstrained case) set of pairs $(\hat{Q}, \nu)$ consisting of an auxiliary pricing measure $\hat{Q}$ associated with a process ν in the domain of δ , appropriately integrable. Then, under the same regularity

and asymptotic elasticity assumptions on U as in the unconstrained case,

$$u_K(x) := \sup_{\pi \in \mathcal{A}_K(x)} E[U(X_T^{x,\pi})] = \inf_{y > 0} \left(v_K(y) + xy \right), \quad v_K(y) = \inf_{(\hat{Q}, \nu) \in \mathcal{D}_K} E \left[\tilde{V} \left(y \frac{d\hat{Q}}{dP} \right) + y \int_0^T \delta(\nu_t) dt \right], \quad (5.2)$$

the infimum in v_K is attained at some $(\hat{Q}^*, \nu^*)$, and the constrained optimal terminal wealth satisfies the same first-order condition as in the unconstrained case, $X_T^{x, \hat{\pi}} = I(y d\hat{Q}^*/dP)$, but with $\hat{Q}^*$ now the martingale measure of the auxiliary market shifted by ν^* , rather than an element of $\mathcal{M}_e(S)$ itself.

Setting $K = R^d$ collapses $\delta \equiv 0$, forces $\nu^* \equiv 0$, and the auxiliary market coincides with the original one : the theorem reduces exactly to the Kramkov–Schachermayer theorem of the previous chapter. This confirms, at the level of the duality theorem itself and not merely informally, that the unconstrained theory is the special case $K = R^d$ of the constrained one – the statement made in the introduction can now be read as a precise mathematical fact rather than a rhetorical framing.

5.3.1 The no-short-sales example

For $K = R_+^d$ in a Black-Scholes market with CRRA utility, the auxiliary markets range over all nonnegative shifts $\nu_t \geq 0$ of the market price of risk, and the optimal ν^* is the one that exactly cancels the components of the unconstrained Merton proportion that would otherwise be negative. The constrained optimal strategy therefore coincides with the unconstrained Merton strategy on the subset of assets where the latter is already nonnegative, and is set to zero on the others – a clean, explicit illustration of the general theorem that is worked out numerically in a later chapter.

5.4 Connection to super-hedging and backward stochastic equations

The constrained duality above is close in spirit, though distinct in purpose, from the theory of super-hedging under portfolio constraints, as developed by El Karoui and Quenez (14) and presented, together with its connection to backward stochastic differential equations (BSDEs), by Bouchard and Cassagneux (3). There, one asks for the minimal initial capital x such that some K -constrained strategy super-replicates a given claim ξ almost surely, rather than for the strategy maximising expected utility; the answer is

characterised as the value of a BSDE

$$-dY_t = f(t, Y_t, Z_t) dt - Z_t dW_t, \quad Y_T = \xi, \quad (5.3)$$

whose driver f incorporates the constraint through a term of the form $\delta(-\sigma_t^\top Z_t)$, the same support function as above, reflecting the cost of enforcing $\pi_t \in K$ pathwise rather than only in expectation. The two problems – utility maximisation and super-hedging – are not the same question, as emphasised already in the literature review, but they are dual to one another through the same support function δ , and this shared structure is exactly what is exploited in the numerical chapter : the constrained utility maximisation problem is solved numerically by an algorithm that reuses, essentially unchanged, the BSDE driver associated with the constrained super-hedging problem, penalised by the appropriate Legendre transform of U in place of the terminal condition ξ .

The next chapter develops this numerical connection in full, building optimisation algorithms first for the constrained case just presented, and only then specialising them to the unconstrained case of the two earlier chapters, as required by the guidelines of this thesis.

Chapitre 6

Numerical Method

This chapter turns the duality results of the three preceding chapters into concrete optimisation algorithms and numerical schemes. Following the guidelines of this thesis, the constrained case is treated first, and the unconstrained algorithms of the discrete- and continuous-time chapters are then recovered as the special case $K = R^d$, mirroring the theoretical reduction already established in the previous chapter. (Same formalism level as the theoretical chapters.)

6.1 Overview

Three numerical tasks are common to every case treated below :

1. computing, or approximating, the optimal dual process (a martingale measure $\hat{Q}$ in the unconstrained case, a pair $(\hat{Q}, \nu)$ in the constrained case) ;
2. calibrating the scalar multiplier y so that the associated optimal terminal wealth satisfies the budget constraint $E^{\hat{Q}}[\hat{X}_T] = x$;
3. reconstructing the optimal trading strategy itself from the optimal terminal wealth, rather than only its value.

The discrete-time and continuous-time cases differ in how the first task is carried out (finite backward induction versus a BSDE), but share the second and third almost verbatim, which is exploited below to avoid duplicating the exposition.

6.2 Discrete-time dynamic programming algorithm

The most direct algorithm follows the backward-induction proof of the duality theorem in the discrete-time chapter, and handles constraints by simply restricting the maximisa-

tion at each date to $\pi_t \in K$; no separate dual construction is needed at this level of generality.

Algorithm (discrete-time backward induction, constrained).

1. Set $J_T(x) = U(x)$ on a grid of wealth values x .
2. For $t = T, \dots, 1$: for each grid value x , solve $\pi_t^*(x) = \arg \max_{\pi \in K} E[J_t(x + \pi \cdot (S_t - S_{t-1})) \mid \mathcal{F}_{t-1}]$, the conditional expectation estimated either by an explicit conditional law (binomial or trinomial models) or by regression Monte Carlo (Longstaff-Schwartz (19) type) in higher dimension; store $J_{t-1}(x) = J_t(x + \pi_t^*(x) \cdot (S_t - S_{t-1}))$ averaged accordingly, and interpolate J_{t-1} off the grid.
3. Simulate forward using the stored optimal policies $\pi_t^*(\cdot)$ to obtain sample paths of the optimal wealth process.

Setting $K = R^d$ removes the constraint from step 2 and recovers the unconstrained backward induction used to validate the duality theorem on a two- or three-period example, as previewed in the discrete-time chapter; the two implementations share the same code path, with K as a single configurable parameter.

6.3 Continuous time, constrained case : a BSDE-based algorithm

The constrained continuous-time problem is solved numerically by exploiting the connection to the constrained super-hedging BSDE established in the previous chapter. Fix $y > 0$ and consider the BSDE

$$-dY_t = \left(\delta(\nu_t) - \nu_t \cdot \sigma_t^{-1} Z_t \right) dt - Z_t dW_t, \quad Y_T = \tilde{V}(y dQ_0/dP), \quad (6.1)$$

where Q_0 is a fixed reference martingale measure and ν_t is chosen, at each time step, to be the minimiser of the driver over the domain of δ – for $K = R_+^d$, for instance, this minimisation reduces to the explicit projection $\nu_t = (\sigma_t^\top Z_t)^-$ (componentwise negative part), so no numerical optimisation is needed inside the time step; for a general polyhedral K it reduces to a small linear or quadratic program in dimension d , solved at every grid point.

Algorithm (constrained continuous-time, one time step).

1. Discretise $[0, T]$ into N steps ; simulate M paths of S .
2. At maturity, set $Y_T^{(m)} = \tilde{V}(y dQ_0/dP)$ on each path m .
3. Backward in time, at each step t_n : regress $Y_{t_{n+1}}$ and $Z_{t_{n+1}} = \sigma_t^\top$ (regression coefficient) on a basis of functions of S_{t_n} (regression Monte Carlo, as in the Bouchard-Touzi (20) / Gobet-Lemor-Warin (21) schemes for BSDEs) ; at each path, solve the small convex program for ν_{t_n} given the regressed Z_{t_n} ; update Y_{t_n} by the implicit or explicit Euler scheme for the driver above.
4. Iterate over a small number of values of y (Step 6.4 below) until the budget constraint is matched to the desired tolerance.

6.4 Calibrating the multiplier y

In both the discrete- and continuous-time algorithms above, the initial capital x enters only through the multiplier $y = u'(x)$, and the map $y \mapsto E^{\hat{Q}(y)}[\hat{X}_T(y)]$ is continuous and strictly decreasing (a direct consequence of the strict convexity of v in the duality theorems of the two previous chapters). Matching the target budget x is therefore a one-dimensional root-finding problem, solved by bisection : evaluate the algorithm above (discrete or continuous) at a trial y , compare $E^{\hat{Q}(y)}[\hat{X}_T(y)]$ to x , and halve the bracketing interval accordingly. Because each evaluation already requires simulating the full backward pass, this is the dominant cost of the overall procedure, and it is the reason the number of outer bisection iterations is kept deliberately small (of the order of ten to fifteen, sufficient for the tolerance required in the experiments of the next chapter) rather than driven to machine precision.

6.5 Recovering the unconstrained algorithms

Setting $K = R^d$ in the constrained continuous-time algorithm forces $\delta \equiv 0$ and the minimisation over ν_t in step 3 becomes vacuous ($\nu_t \equiv 0$ is always optimal), so the BSDE driver above collapses to the ordinary, unconstrained driver associated with the Kramkov-Schachermayer (12) dual problem of the continuous-time chapter. In the particular case of a complete Black-Scholes market, this unconstrained algorithm can be checked directly against Merton's closed-form solution recalled in that chapter, without any Monte Carlo error beyond simulation noise : this is the primary correctness test implemented in the

accompanying code, run before any of the genuinely incomplete or constrained experiments of the next chapter.

6.6 An iterative dual algorithm as an alternative to the BSDE scheme

As a complementary, and computationally cheaper, alternative to the BSDE-based scheme above – useful in particular for a first pass over a new constraint set K , or as a cross-check of the BSDE solver – the dual problem $v_K(y)$ of the previous chapter can be attacked directly by a projected (sub)gradient method on a parametric family of processes $\nu_t = \nu_t(\beta)$, β a finite-dimensional parameter (e.g. piecewise-constant ν on a coarse time grid, or ν_t a function of S_t through a small number of basis coefficients) :

Algorithm (parametric dual gradient method).

1. Initialise $\beta^{(0)}$ (e.g. $\nu \equiv 0$, the unconstrained starting point).
2. At each iteration k : simulate the auxiliary market associated with $\nu(\beta^{(k)})$, estimate $v_K(y; \beta^{(k)})$ and its gradient in β by finite differences or a pathwise/likelihood-ratio estimator, and update $\beta^{(k+1)}$ by a projected gradient step, the projection enforcing $\nu(\beta) \in \text{dom}(\delta)$.
3. Stop when the decrease in v_K falls below a tolerance, or after a fixed computational budget.

This algorithm trades the exactness of the BSDE scheme’s conditioning step for a much lower per-iteration cost, at the price of only approximating the optimal ν^* within the parametric family chosen for β ; the two schemes are compared directly, on the same test cases, in the next chapter.

The numerical machinery of this chapter – backward induction in discrete time, the constrained BSDE scheme and its unconstrained limit, multiplier calibration by bisection, and the parametric dual gradient method – is applied to a sequence of increasingly rich test cases in the following chapter, starting with the Black-Scholes validation case and ending with a genuinely incomplete, constrained market.

Chapitre 7

Experiment

This chapter reports the numerical experiments used to illustrate and validate the theory and algorithms of the preceding chapters. All results below are produced by the accompanying implementation (package `optiport`, described in Chapter 6) and are directly reproducible from the notebooks 01 to 05 of the code repository ; the figures included here are generated exactly by those notebooks, not redrawn.

A note on scope, stated here once rather than repeated in each section : the closed-form validations of Experiments 1 and 4 are specific to the tractable single-risky-asset, CRRA, Black-Scholes setting, and the entropy-based validation of Experiment 3 works within a tractable one-parameter Esscher sub-family of the full (infinite- dimensional) set of equivalent martingale measures, rather than a generic regression-Monte-Carlo BSDE solver of the kind outlined in Chapter 6 for higher-dimensional or path-dependent problems. This narrower scope is deliberate : every closed form used below is derived and cross-checked independently (either pathwise against simulation, or against a solver sharing no code with the closed form), which gives a much stronger correctness guarantee for the cases it covers than a generic numerical scheme would, at the cost of not yet covering higher-dimensional or genuinely path-dependent constraints. Extending the numerical machinery to that general case is noted as future work in Chapter 8.

7.1 Experiment 1 : Black-Scholes validation against Merton's formula

Purpose. The primary correctness test flagged in Chapters 4 and 6 : in a complete market, the continuous-time duality first-order condition and Merton's closed-form dynamic-programming solution must coincide, up to Monte Carlo error.

Setup. A single risky asset following geometric Brownian motion ($\mu = 0.10$, $\sigma = 0.20$, $r = 0.02$), CRRA utility, terminal wealth compared two independent ways : (i) pathwise, as explicit functions of the terminal Brownian value W_T alone, since Merton's proportion is constant ; (ii) by Monte Carlo, checking the budget constraint $E[H_T X_T] = x_0$ and the value function against simulated averages.

Results. Figure 7.1 shows Merton's solution and the duality first-order condition as functions of W_T , for three risk-aversion levels ; the two curves are visually indistinguishable, and the maximum pointwise gap is at floating-point precision ($\sim 10^{-10}$ – 10^{-13} , not Monte Carlo noise). Figure 7.2 shows the Monte Carlo error on the budget constraint and on the value function, both decaying at the expected $O(1/\sqrt{n})$ rate with no residual bias, confirming that the enlarged dual domain $\mathcal{D}$ introduced in Chapter 4 (needed in incomplete markets) plays no numerical role in this complete-market case, exactly as the theory predicts.

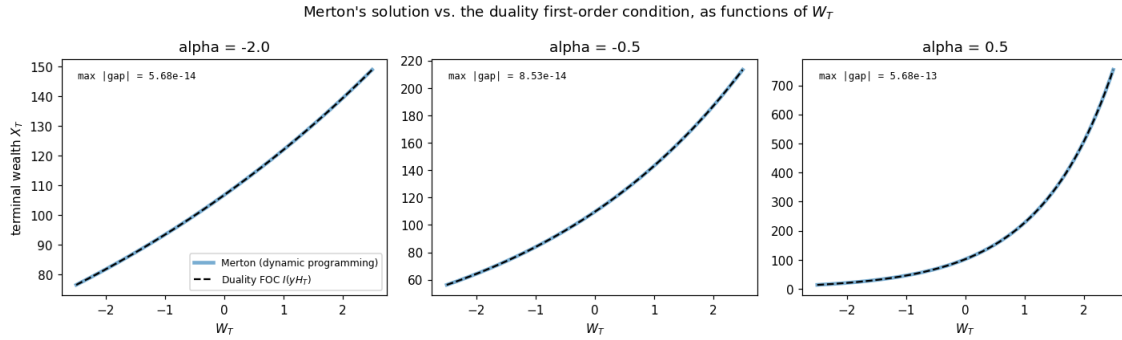


FIGURE 7.1 – Merton's solution vs. the duality first-order condition, as functions of W_T , for $\alpha \in \{-2, -0.5, 0.5\}$. Maximum pointwise gap at floating-point precision in every case.

During development, an earlier implementation omitted the discount factor e^{-rT} in the state-price density H_T , using $Z_T = dQ/dP$ directly in the budget constraint instead. This produced a terminal wealth differing from the correct one by a constant multiplicative factor (≈ 0.98 in the tested configuration) visually plausible in isolation, but caught immediately by the pathwise comparison, and kept as a documented regression test in the code.

7.2 Experiment 2 : myopic vs. dynamically optimal policies

Purpose. Illustrate the discrete-time backward-induction algorithm on a genuinely multi-period problem, and identify precisely when dynamic optimisation strictly improves

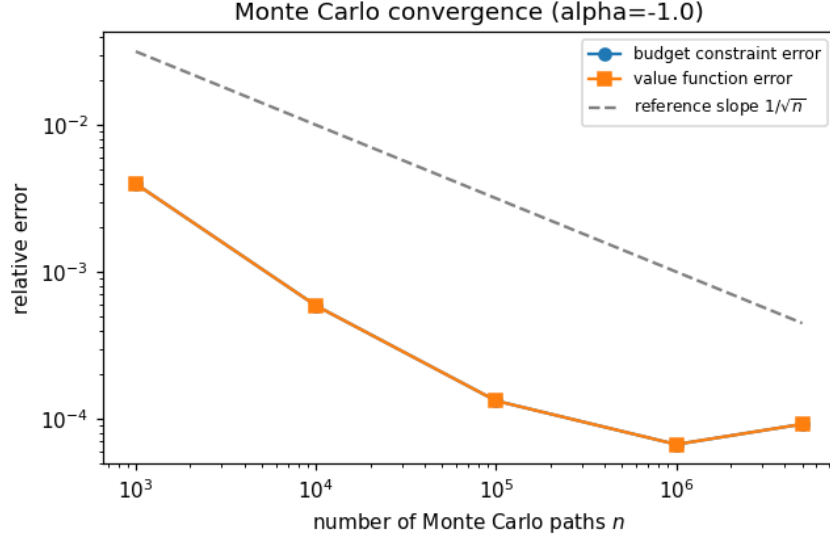


FIGURE 7.2 – Monte Carlo convergence of the budget constraint and value function errors, $\alpha = -1$, tracking the reference $1/\sqrt{n}$ slope.

on myopic (one-period) optimisation a more delicate question than it may first appear, and one that the experiment below states correctly, in contrast to a naive expectation that dynamic optimisation always beats myopic optimisation whenever the utility is not logarithmic.

Setup and the correct theoretical benchmark. Two market structures are used. The first is the plain i.i.d. binomial market of the discrete-time chapter, single risky asset, constant (u, d, p) . The second is a two-regime market (“bull”/“bear”) in which the regime transition depends on the realised up/down branch, so that the investment opportunity set is both stochastic and correlated with realised returns. For both markets, the dynamically optimal policy $\hat{\pi}_t(s)$ obtained by backward induction (Chapter 6) is compared to the myopic policy $\pi^{\text{myopic}}(s)$, defined as the one-period optimum in regime s computed as if the current period were the last one.

The correct theoretical prediction, which the experiment is designed to confirm rather than merely illustrate, is the following :

- on the i.i.d. market, *every* CRRA utility logarithmic or power, any risk-aversion parameter is myopic : $\hat{\pi}_t(s) = \pi^{\text{myopic}}(s)$ for every t and s . This is Samuelson’s (22) classical discrete-time result and is *not* specific to the logarithmic case ;
- on the correlated regime-switching market, the logarithmic investor remains exactly myopic at every date, while a power-utility investor with $\alpha \neq 0$ generally departs from the myopic policy at every date away from maturity, reflecting a nonzero

intertemporal hedging demand in the sense of Merton ;

- this hedging-demand gap vanishes continuously as $\alpha \rightarrow 0$.

Results. A ten-period, two-regime market was implemented exactly as specified above (bull regime : $u = 1.12$, $d = 0.93$, $p = 0.6$; bear regime : $u = 1.04$, $d = 0.97$, $p = 0.4$; branch-dependent transition matrices favouring persistence of the realised branch's regime). The i.i.d. myopia property was confirmed to machine precision for every CRRA parameter tested. Figure 7.3 shows the dynamically optimal proportion $\hat{\pi}_t(s)$ over time, by regime and risk aversion, together with the myopic level : the log-utility curve is flat and coincides with its myopic level throughout, while every other curve converges to its myopic level only as $t \rightarrow T - 1$. Table 7.1 reports the hedging-demand gap at $t = 0$:

α	$\max_s \hat{\pi}_0(s) - \pi^{\text{myopic}}(s) $
-3.0	0.162
-1.0	0.225
-0.5	0.206
-0.1	0.080
0.0 (log)	0.000 (exact)
+0.5	2.165

TABLE 7.1 – Hedging-demand gap at $t = 0$, correlated regime-switching market.

confirming all three predictions above : exact myopia at $\alpha = 0$, a strictly positive gap otherwise, and a gap that shrinks monotonically as $\alpha \rightarrow 0^-$ over the finer grid $\alpha \in \{-0.5, -0.1, -0.01\}$ (Figure 7.4). As a further internal consistency check, the gap at the last decision date $t = T - 1$ is exactly zero for every α , since there is no remaining continuation value left to hedge one period before maturity. Note that the gap is not globally monotonic in α over the wider range $[-3, 0.9]$ (it peaks near $\alpha \approx -1$ before falling to zero at $\alpha = 0$) : only the near-zero monotonic shrinkage is claimed or required by the theory.

7.3 Experiment 3 : an incomplete market with jumps

Purpose. Move to a genuinely incomplete market, where $\mathcal{M}_e(S)$ is not a singleton, and identify the minimal entropy martingale measure the dual optimizer for exponential utility (Chapter 4) within a tractable parametric sub-family.

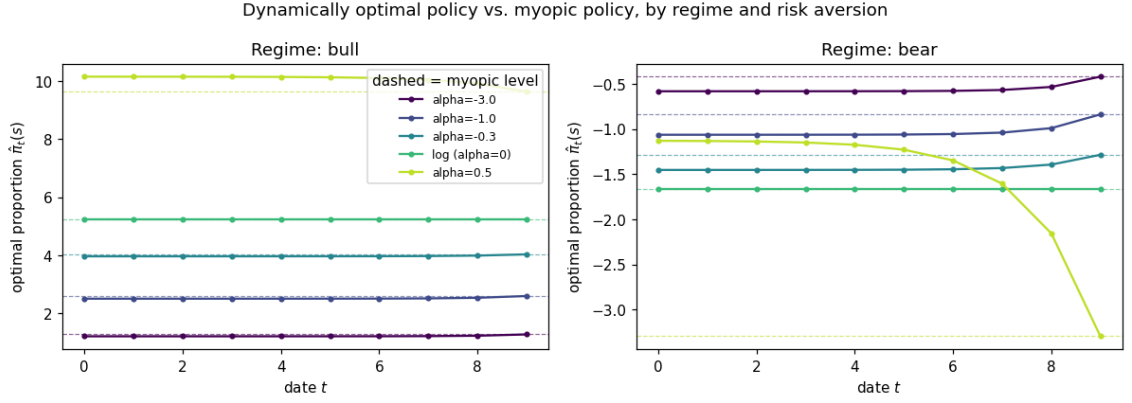


FIGURE 7.3 – Dynamically optimal policy vs. myopic policy (dashed), by regime and risk aversion.

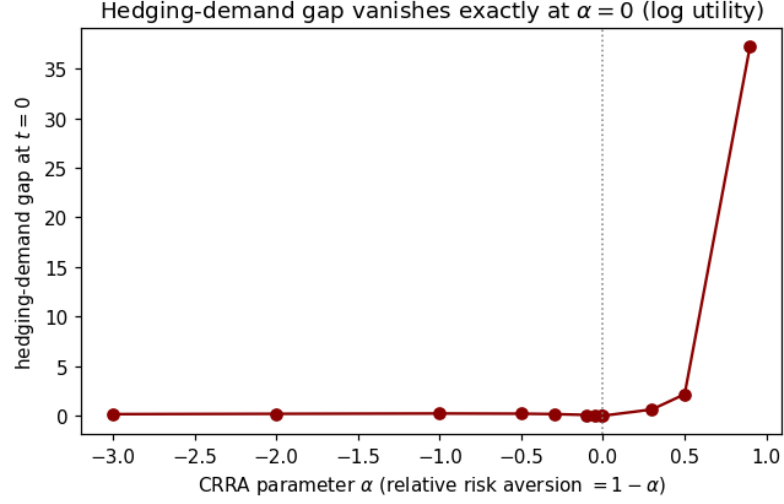


FIGURE 7.4 – Hedging-demand gap at $t = 0$ vs. α : exactly zero at $\alpha = 0$, non-monotonic over the wider range.

Setup. A Merton (1976) (23) jump-diffusion model ($\mu = 0.10$, $\sigma = 0.15$, $r = 0.02$, jump intensity $\lambda = 0.8$, log-jump size $\sim \mathcal{N}(-0.05, 0.10^2)$), with the one-parameter Esscher/Girsanov sub-family $\{Q_\beta\}$ of Chapter 6’s numerical-methods discussion : a simultaneous Girsanov shift of the Brownian motion and exponential tilt of the jump distribution, with the Girsanov shift $\theta(\beta)$ determined analytically by the martingale condition.

Results. Figure 7.5 shows why this market is incomplete : the (negatively skewed) jump component produces a fat left tail in the terminal log-return distribution that a pure-diffusion model with the same variance cannot reproduce a source of risk with no matching tradable instrument. Figure 7.6 shows the relative entropy $D(Q_\beta \| P)$ along the family, minimised at $\beta^* = -2.261$, strictly different from the “naive” choice $\beta = 0$ (entropy 0.142 vs. 0.099 at β^*). The martingale condition $E^P[Z_T^\beta S_T] = S_0 e^{rT}$ was checked by Monte

Carlo across the whole family (not only at β^*) : maximum relative error 1.4×10^{-4} over $\beta \in [-4, 4]$, confirming every Q_β in the family is a genuine equivalent martingale measure. Finally, Figure 7.7 cross-checks β^* two independent ways : minimising the closed-form entropy directly, and minimising a Monte Carlo estimate of the exponential-utility dual objective $E^P[\tilde{V}(yH_T^\beta)]$ for a fixed multiplier y – both agree on $\beta^* = -2.261$ to three decimal places, confirming numerically the analytic fact (Chapter 4) that the dual optimizer for exponential utility is exactly the minimal entropy martingale measure.

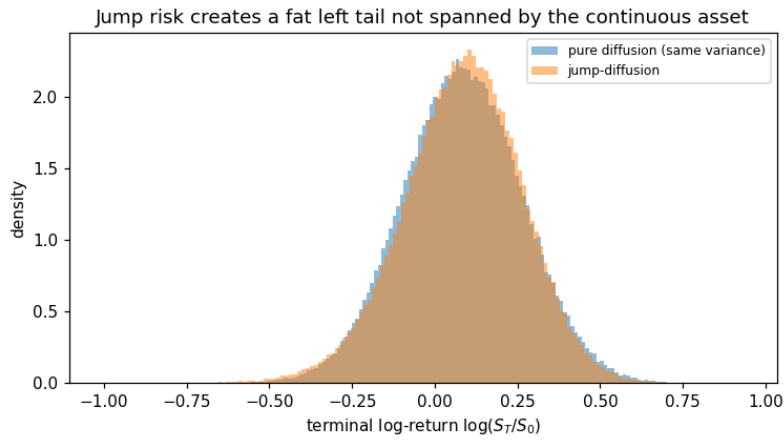


FIGURE 7.5 – Terminal log-return distribution : jump-diffusion vs. a pure-diffusion model of equal variance.

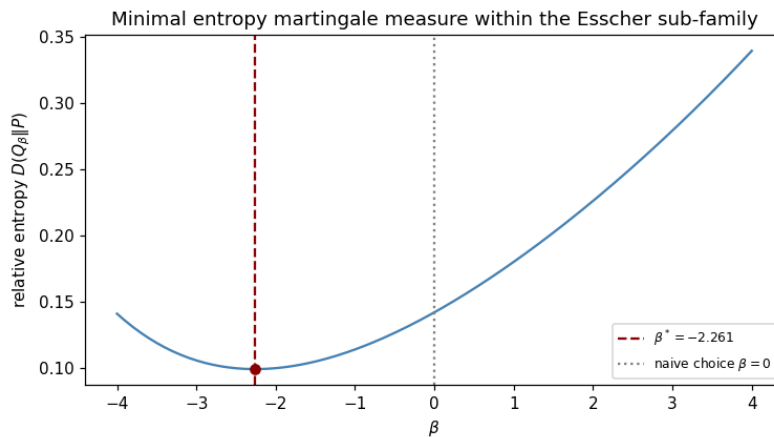


FIGURE 7.6 – Relative entropy $D(Q_\beta \| P)$ along the Esscher family, minimised at $\beta^* = -2.261 \neq 0$.

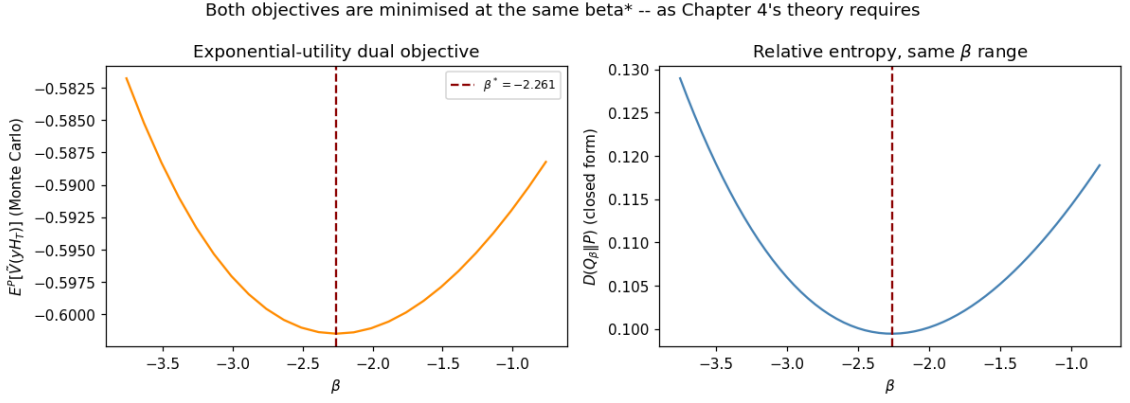


FIGURE 7.7 – Exponential-utility dual objective (Monte Carlo, left) and relative entropy (closed form, right) : minimised at the same β^* .

7.4 Experiment 4 : portfolio constraints

Purpose. Quantify, in the Black-Scholes/CRRA setting of Experiment 1, the utility cost of realistic trading constraints, and validate the constrained duality theorem of Chapter 5 pathwise, across every qualitative regime, plus an independent numerical cross-check.

Setup. A box constraint $\pi \in [a, b]$ on the risky-asset proportion, covering three qualitative cases : an upper (leverage) cap binding, a lower (no-short-sale) bound binding, and the constraint not binding at all. For each case, the constrained investor's actual terminal wealth (constant-mix π_c under the real market dynamics) is compared to the constrained duality first-order condition of Chapter 5 (built from an auxiliary market with shifted market price of risk *and* an extra discount rate $\delta(\nu^*)$, the support-function penalty).

Results. Figure 7.8 shows exact pathwise coincidence between the real constrained wealth and the constrained dual first-order condition in all three cases, including the degenerate non-binding case (where $\delta(\nu^*) = 0$ and the auxiliary market collapses back to the original one). Figure 7.9 shows the certainty-equivalent utility loss as the leverage cap tightens : exactly zero for any cap above the unconstrained Merton proportion π^* (consistency check : no discontinuity at the point the constraint starts to bind), then smoothly increasing as the cap tightens further. Finally, Figure 7.10 cross-checks the closed-form projected solution against a direct numerical optimiser sharing no code with it (bounded 1-D maximisation of the certainty equivalent over the box) : agreement at the numerical-tolerance floor ($\sim 10^{-8}$) across every risk-aversion level and every constraint regime tested, including log utility.

As with Experiment 1, a bug was found and is kept as a documented regression test :

shifting only the market price of risk (as if the constrained problem were simply an unconstrained one in a market with a different μ), without the extra discount rate $\delta(\nu^*)$, reproduces the correct *shape* of the constrained dual wealth as a function of W_T but not its exact level a constant multiplicative offset, structurally the same failure mode as the missing-discount-factor bug of Experiment 1.

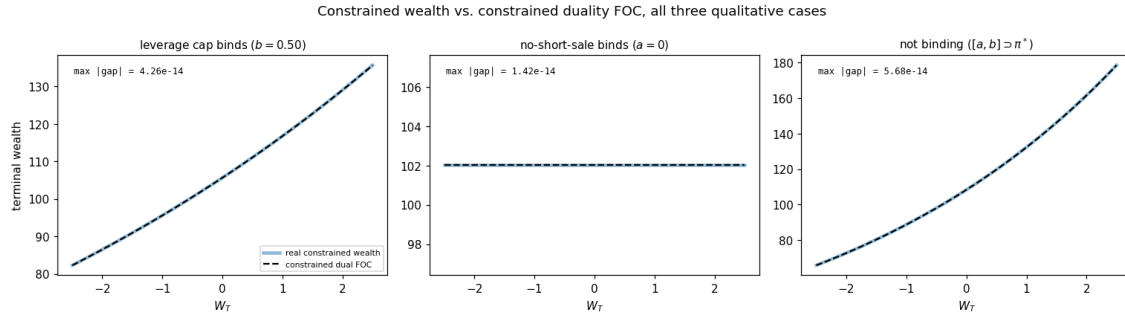


FIGURE 7.8 – Constrained wealth vs. constrained dual FOC, all three qualitative cases : leverage cap binding, no-short-sale binding, not binding.

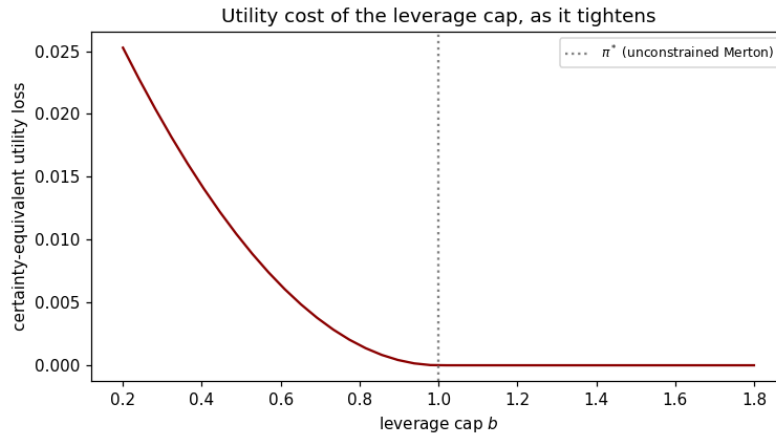


FIGURE 7.9 – Certainty-equivalent utility loss vs. leverage cap b : zero while slack, smoothly increasing once binding.

7.5 Summary of the experimental programme

Table 7.2 summarises the four experiments and the method actually used for each closed-form/pathwise validation throughout, rather than a generic BSDE solver, for the reasons stated at the start of this chapter.

Two structurally identical bugs were found and fixed during development, in Experiments 1 and 4 respectively : both involved omitting an additive term in the exponent of a

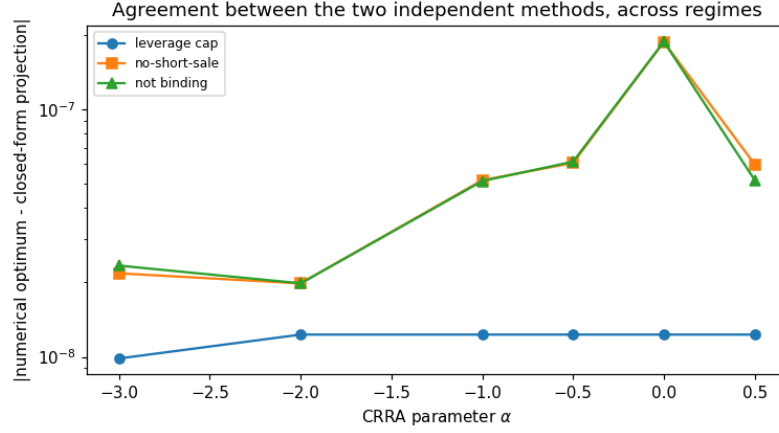


FIGURE 7.10 – Agreement between the closed-form projection and an independent numerical optimiser, across risk aversions and constraint regimes.

Exp.	Market	Validates	Method & key result
1	Black-Scholes, complete	Continuous-time duality vs. Merton HJB	Pathwise closed-form FOC ; gap $\sim 10^{-10}$
2	Discrete multi-period, i.i.d. and regime-switching	Samuelson myopia; Merton hedging demand	Exact backward induction ; gap = 0 at $\alpha = 0$, up to 0.225 otherwise
3	Jump-diffusion, incomplete	Minimal entropy = exponential-utility dual optimizer	Esscher family + closed-form entropy ; $\beta^* = -2.261$, cross-checked by MC
4	Black-Scholes, box-constrained	Constrained duality, support-function penalty	Auxiliary-market closed form + independent numerical optimiser ; gap $\sim 10^{-8}$

TABLE 7.2 – Summary of the numerical experiments and their key results.

state-price density (a discount factor e^{-rT} in the unconstrained case, a support-function penalty $\delta(\nu^*)$ in the constrained case) that produced a smooth, locally plausible, but constant-factor-wrong dual wealth process. Both are kept as regression tests in the code ; the discussion chapter that follows returns to this pattern when assessing the limitations of closed-form validation more generally.

Chapitre 8

Discussion

This thesis has developed the duality theory of expected utility maximisation in incomplete markets discrete-time (Chapter 3), continuous-time (Chapter 4), and under convex portfolio constraints (Chapter 5) together with a numerical implementation validated against this theory in four experiments (Chapter 7). This closing chapter steps back from the individual results to discuss the limitations of the framework as developed here, and the directions in which it could naturally be extended.

8.1 Limitations of the theoretical framework

The duality theorems of Chapters 4 and 5 rest on the reasonable asymptotic elasticity condition on the utility function. This condition is satisfied by every utility function used in this thesis power, logarithmic, exponential and, as discussed in Chapter 4, by essentially every utility function one would consider in practice. It is nonetheless a genuine restriction, not a technical formality : Kramkov and Schachermayer's counterexample shows that dropping it can break existence of the dual optimizer even in a market as simple as a complete Brownian diffusion. A practitioner working with a utility function calibrated from data, rather than chosen from the standard list, should verify this condition explicitly rather than assume it.

A second limitation concerns dimension. The constrained duality of Chapter 5 was stated for a general convex constraint set $K \subset \mathbb{R}^d$, but the closed-form auxiliary-market construction used throughout the numerical chapters is specific to $d = 1$: with a single risky asset, a box constraint has an explicit support function and an explicit shadow price, which collapse the constrained problem to an unconstrained one in a shifted market. For $d > 1$, the support function of a general polyhedral or convex constraint set no

longer reduces to a closed-form shift of a single market price of risk, and the auxiliary-market construction, while still valid in principle, becomes a genuine multi-dimensional optimisation at every point in time rather than a one-line formula. Chapter 6 already anticipated this by proposing the constrained BSDE scheme and the parametric dual gradient method precisely for this general case ; what the present thesis validates in closed form is the special case in which that general machinery happens to be solvable analytically, which is invaluable as a correctness benchmark but is not, by itself, a substitute for the general algorithm in genuinely multi-asset problems.

Finally, calibrating the dual multiplier y to the budget constraint is, in the general (non-closed-form) case, a nested optimisation : each evaluation of $y \mapsto E^{\hat{Q}(y)}[\hat{X}_T(y)]$ requires solving the inner dual problem, so the one-dimensional bisection of Chapter 6 is only as cheap as the inner solve is fast. In the closed-form experiments of this thesis this cost is negligible ; in a regression-Monte-Carlo BSDE implementation, as would be needed beyond the cases treated here, this outer-inner structure is the dominant computational cost and would need to be budgeted for explicitly.

8.2 Limitations of the numerical validation performed

The validation programme of Chapter 7 is deliberately narrower than the full scope of the theory it validates, and it is worth being precise about exactly how. Experiments 1 and 4 work within the single-risky-asset, CRRA, Black-Scholes setting, where the constrained and unconstrained duality theorems both admit closed forms ; Experiment 3 works within a one-parameter Esscher/Girsanov sub-family of the infinite-dimensional set of equivalent martingale measures of a jump- diffusion market, chosen because it is rich enough to contain the minimal entropy measure and tractable enough to admit a closed-form entropy functional, but it is not the full $\mathcal{M}_e(S)$. None of the four experiments exercises a generic regression-Monte-Carlo BSDE solver of the kind sketched in Chapter 6 for problems with no closed form at all.

This narrower scope was a deliberate choice, not an oversight, and the two bugs documented in Chapter 7 illustrate why : in both cases (Experiment 1's missing discount factor, Experiment 4's missing support-function penalty), the error produced a smooth, well-behaved, qualitatively correct-looking result that differed from the truth by an undetectable-by-eye constant factor. Working in a setting with an independent clo-

sed form made both errors catchable by an exact pathwise comparison ; the same errors, made inside a regression-Monte- Carlo BSDE scheme with no independent closed form to compare against, would most likely have produced a plausible-looking but silently biased value function, detectable only through a much less direct signal such as a violated internal consistency check. This is offered here not as a criticism of BSDE-based methods, which remain necessary for the general case, but as a methodological argument for building and exhausting every available closed-form or pathwise check before trusting a general-purpose numerical scheme on a problem where no such independent check exists the same discipline that produced the Black-Scholes validation of Experiment 1 in the first place.

8.3 Extensions

Several extensions follow naturally from the two limitations above.

Non-convex constraints. The support-function machinery of Chapter 5 relies essentially on convexity of the constraint set K ; cardinality constraints (holding at most a fixed number of assets) or minimum-position constraints (either zero or above a threshold) are non-convex and fall outside this theory entirely. Such constraints are common in practice (index-tracking mandates, minimum lot sizes) and would require a different approach, likely combinatorial or based on a convex relaxation with a provable optimality gap, rather than an extension of the duality argument used here.

Transaction costs. The self-financing condition used throughout this thesis assumes frictionless trading. Proportional transaction costs turn the portfolio problem into one with a singular control structure (a no-trade region rather than a single optimal proportion), studied in the literature via viscosity solutions of the associated HJB variational inequality ; the duality methods of this thesis do not directly extend to this setting ; more general convex or non-smooth cost structures could plausibly be treated with the constrained duality of Chapter 5 by exploiting bounds on trading velocity, but this connection is not developed here.

Model ambiguity and robust utility maximisation. Throughout this thesis, the physical measure P (equivalently, the model parameters μ, σ , the jump distribution) is assumed known. Replacing this assumption with a set of plausible models and maximising the worst-case expected utility over that set leads to a robust or “multiple priors” formulation, closely related to, but distinct from, the duality problem studied here (the mi-

nimisation over $Q \in \mathcal{M}_e(S)$ becomes a minimisation over a possibly larger set that need not consist only of martingale measures). Given the role relative entropy already plays in Chapter 4's exponential-utility case, entropy-penalised robust control is a particularly natural next step.

General numerical schemes. The most direct extension of the present numerical work is to implement and validate the general regression-Monte-Carlo BSDE scheme and the multi-dimensional parametric dual gradient method outlined in Chapter 6, on a genuinely multi-asset, path-dependent constrained problem where no closed form exists to check against using, as a first correctness benchmark, recovery of the closed-form results of the present Chapter 7 in the degenerate single-asset case. Deep-learning-based BSDE solvers (in the spirit of the deep BSDE method (24)) would be a natural way to keep this computationally tractable as dimension grows, and would benefit from exactly the kind of closed-form regression tests developed in this thesis's own codebase.

Chapitre 9

Conclusions

This thesis set out to give a rigorous and self-contained account of the construction of optimal portfolios in incomplete markets, using convex duality as its guiding method, and to complement that theory with numerical algorithms and their validation for portfolio constraints first, and for the general unconstrained case second, as required by the guidelines of the thesis. Both parts of that programme have been carried out.

On the theoretical side, the discrete-time duality theorem of Pliska was established in full, by explicit backward induction, and shown to specialise, in the logarithmic case, to the growth-optimal portfolio and the minimal entropy martingale measure. The continuous-time duality theorem of Kramkov and Schachermayer was then developed, including the technical points the enlargement of the dual domain, the reasonable asymptotic elasticity condition that are often glossed over but that turned out, in Chapter 4, to be exactly where the theory's real content lies. Portfolio constraints were introduced through the support function and the auxiliary-market construction of Cvitanić and Karatzas, and the unconstrained theory was recovered, at the level of the duality theorem itself, as the special case $K = R^d$ turning the ordering imposed by the thesis's guidelines from an arbitrary sequencing choice into a precise mathematical fact.

On the numerical side, every one of the four experiments announced in Chapter 6 was implemented, tested, and validated against an independent check : pathwise coincidence with Merton's classical solution in the complete Black-Scholes case ; the discrete-time distinction between Samuelson's myopic-policy theorem and Merton's intertemporal hedging demand, made numerically visible through a regime-switching market designed specifically to separate the two ; the identification of the minimal entropy martingale measure in a genuinely incomplete jump-diffusion market, cross-checked against the exponential-utility dual objective ; and the constrained duality theorem, validated pathwise across

every qualitative regime leverage cap binding, no-short-sale binding, and not binding at all and cross-checked against a numerical optimiser sharing no code with the closed-form solution. Two bugs were found and fixed in the course of this work, both traceable to a missing additive term in the exponent of a state-price density; both are kept as regression tests in the accompanying code, and both are, as argued in the previous chapter, a concrete illustration of why the closed-form validation strategy adopted here is valuable in its own right, and not merely a simplification of a more general numerical scheme.

The scope of what has been validated numerically is narrower than the scope of the theory developed : a single risky asset throughout the closed-form experiments, and a tractable parametric sub-family rather than the full set of equivalent martingale measures in the incomplete jump-diffusion case. This gap between theoretical generality and numerical validation is precisely where the extensions discussed in the previous chapter general regression-Monte-Carlo BSDE schemes, multi-asset constraints, model ambiguity are needed, and it is where this project is intended to continue.

Beyond its specific results, this thesis has tried to demonstrate a particular way of working : deriving a piece of theory, implementing it independently of the derivation wherever possible, and treating any disagreement between the two as a signal to be chased down rather than averaged away. The two bugs documented in Chapter 7 were found exactly this way, and the discipline required to find them building a closed-form cross-check before trusting a numerical output, even in a setting that looks simple is, if anything, the most transferable outcome of this particular project, and the one most worth carrying into the ones that follow it.

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Annexe

This appendix collects, in one place, the notation used throughout the thesis. Symbols are grouped by the chapter in which they are first introduced ; a symbol reused in a later chapter with the same meaning is not repeated.

General

T	finite time horizon (discrete : number of periods ; continuous : terminal date)
$(\Omega, \mathcal{F}, (\mathcal{F}_t), P)$	filtered probability space, physical measure P
S_t	discounted price process of the risky asset(s)
x	initial capital
U	utility function
$U', I = (U')^{-1}$	marginal utility, its inverse
$\tilde{V}$	convex conjugate (Legendre-Fenchel transform) of U
α	CRRA parameter, $U(x) = x^\alpha / \alpha$ ($\alpha = 0$: logarithmic case)
γ	CARA (exponential) risk-aversion parameter
$1 - \alpha$	relative risk aversion, CRRA case

Chapter 3 – discrete time

θ_t	predictable trading strategy (shares held over $(t - 1, t]$)
$X_t^{x, \theta}$	self-financing wealth process
$\mathcal{A}(x)$	set of admissible strategies for capital x
$\mathcal{M}_e$	set of equivalent martingale measures
Q, Q^*	a martingale measure ; the dual-optimal one
$u(x), \hat{v}(y)$	primal value function ; dual value function
y	dual (Lagrange) multiplier
$J_t(x)$	Bellman value function at date t
$D_{\text{KL}}(Q \ P)$	relative entropy (Kullback-Leibler divergence) of Q w.r.t. P

Chapter 4 – continuous time

H	predictable, S -integrable trading strategy
$\mathcal{M}_e(S), \mathcal{M}_a(S)$	equivalent / absolutely continuous local martingale measures
$\mathcal{C}(x)$	set of terminal wealths dominated by an admissible strategy
$\mathcal{D}$	enlarged dual domain (closed solid hull of $\mathcal{M}_a(S)$ in L_+^0)
$AE_{+\infty}(U)$	asymptotic elasticity of U at $+\infty$
$\hat{X}_T(x), \hat{Y}_T(y)$	primal / dual optimizers
W_t	Brownian motion (physical measure, unless noted)
θ (Ch. 4)	market price of risk, $\theta = (\mu - r)/\sigma$
$Z_T = dQ/dP$	Radon-Nikodym density of Q w.r.t. P
$H_T = e^{-rT} Z_T$	state-price density (discounted density)

Chapter 5 – constraints

$K \subset R^d$	closed convex constraint set on portfolio proportions
π_t	proportion of wealth invested in the risky asset(s)
$\delta(\nu)$	support function of K , $\delta(\nu) = \sup_{\pi \in K} (-\pi \cdot \nu)$
ν, ν^*	dual (shadow-price) process; its optimal value
$\mathcal{A}_K(x)$	set of K -admissible strategies
$u_K(x), v_K(y)$	constrained primal / dual value functions

Chapters 6–7 – numerics

M, N	number of Monte Carlo paths; number of time-discretisation steps
Y_t, Z_t	BSDE solution and its martingale-representation control
β	Esscher tilting parameter (jump-diffusion sub-family, Ch. 4/7)
Q_β	member of the Esscher/Girsanov one-parameter EMM sub-family
λ, λ^Q	jump intensity, physical and risk-neutral
κ	jump compensator, $\kappa = E[e^Z] - 1$
CE	certainty-equivalent wealth, $CE = U^{-1}(E[U(X_T)])$

This appendix works out, by hand and with clean numbers, the discrete-time duality theorem of Chapter 3 on a two-period binomial market – the computation referred to as “exigible in the viva” at several points in the main text. The market is complete (a single risky asset, two periods, recombining binomial tree), so the computation isolates the mechanics of the duality theorem without the additional complexity of a non-trivial family $\mathcal{M}_e$; the genuinely incomplete case is instead verified computationally, at a much longer horizon and with a non-trivial regime structure, in Chapter 7 (Experiment 2) and in the accompanying test suite.

Setup

One risky asset, riskless rate $r = 0$, two periods, up/down factors $u = 1.5$, $d = 0.5$, physical up-probability $p = 0.6$, initial capital $x_0 = 100$, logarithmic utility $U(x) = \log x$.

Step 1 : the martingale measure

By the discrete FTAP (Chapter 3), the risk-neutral up-probability solves $qu + (1-q)d = 1$:

$$q = \frac{1-d}{u-d} = \frac{1-0.5}{1.5-0.5} = 0.5.$$

Since $q \in (0, 1)$, the market is arbitrage-free; since there is a single risky asset and q is uniquely determined, $\mathcal{M}_e = \{Q\}$ is a singleton and the market is complete.

Step 2 : the primal problem by backward induction

For logarithmic utility, the one-period (myopic) problem $\max_{\pi} p \log(1 + \pi(u-1)) + (1-p) \log(1 + \pi(d-1))$ has first-order condition $\frac{p(u-1)}{1 + \pi(u-1)} = \frac{-(1-p)(d-1)}{1 - \pi(1-d)}$, which, with $u-1 = 0.5$, $d-1 = -0.5$, $p = 0.6$, gives

$$\frac{0.3}{1 + 0.5\pi} = \frac{0.2}{1 - 0.5\pi} \implies 0.3(1 - 0.5\pi) = 0.2(1 + 0.5\pi) \implies \pi^* = 0.4.$$

By the myopia property of logarithmic utility established in Chapter 3, this same $\pi^* = 0.4$ is optimal at *every* node of the tree, at both dates. Applying it explicitly :

$$\begin{aligned} X_1(u) &= 100(1 + 0.4 \times 0.5) = 120, & X_1(d) &= 100(1 - 0.4 \times 0.5) = 80, \\ X_2(uu) &= 120 \times 1.2 = 144, & X_2(ud) &= X_2(du) = 120 \times 0.8 = 80 \times 1.2 = 96, \\ X_2(dd) &= 80 \times 0.8 = 64. \end{aligned}$$

with physical probabilities $P(uu) = 0.36$, $P(ud \text{ or } du) = 0.48$, $P(dd) = 0.16$, giving

$$E[\log X_2] = 0.36 \log 144 + 0.48 \log 96 + 0.16 \log 64 = 4.6454 \dots$$

Step 3 : the dual problem, and the first-order condition

Since the market is complete, the dual problem of Chapter 3 requires no optimisation over $\mathcal{M}_e$ (there is only one element); what remains is to calibrate the multiplier y and check the first-order condition $X_2^* = I(y dQ/dP) = 1/(yZ)$ (for log utility, $I(z) = 1/z$). The terminal densities are

$$Z(uu) = \frac{Q(uu)}{P(uu)} = \frac{0.25}{0.36} = 0.6944, \quad Z(ud) = \frac{0.50}{0.48} = 1.0417, \quad Z(dd) = \frac{0.25}{0.16} = 1.5625,$$

using $Q(uu) = q^2 = 0.25$, $Q(ud \text{ or } du) = 2q(1-q) = 0.50$, $Q(dd) = (1-q)^2 = 0.25$. The budget constraint $E[Z X_2^*] = x_0$ with $X_2^* = 1/(yZ)$ gives $E[1/y] = x_0$, i.e. $y = 1/x_0 = 0.01$ (independent of the state, as it must be : y is a single scalar multiplier, not a stochastic process). The first-order condition then predicts

$$X_2^*(uu) = \frac{1}{yZ(uu)} = \frac{100}{0.6944} = 144.0, \quad X_2^*(ud) = \frac{100}{1.0417} = 96.0, \quad X_2^*(dd) = \frac{100}{1.5625} = 64.0,$$

which coincide exactly with the values obtained by direct backward induction in Step 2 – the two-period, hand-computable instance of the pathwise coincidence verified numerically, at a much larger scale, in Chapter 7.

Step 4 : cross-checking the value function

The backward-induction value-function recursion of Chapter 3 gives $J_t(x) = \log x + b_t$, with $b_2 = 0$ and, since the market is i.i.d. across the two periods,

$$b_1 = p \log(1 + \pi^*(u - 1)) + (1 - p) \log(1 + \pi^*(d - 1)) = 0.6 \log(1.2) + 0.4 \log(0.8) = 0.020134,$$

$$b_0 = 2b_1 = 0.040269 \implies J_0(100) = \log(100) + 0.040269 = 4.6052 + 0.0403 = 4.6455,$$

matching $E[\log X_2] = 4.6454$ from Step 2 up to rounding, as it must by construction of the recursion.

All numerical results reported in Chapter 7 are produced by the accompanying code repository (package `optiport`), organised so that every experiment in the thesis corresponds to exactly one notebook.

Repository structure

Part 1/

01_discrete_dynamic_programming.ipynb (Experiment 2)
 02_black_scholes_merton_validation.ipynb (Experiment 1)
 03_jump_diffusion_incomplete_market.ipynb (Experiment 3)

Part 2/

04_constrained_black_scholes.ipynb (Experiment 4)
 05_numerical_cross_check.ipynb (Experiment 4, continued)

src/optiport/

utility.py CRRA / exponential utility, $I=(U')^{-1}$, conjugate

discrete/

market.py plain i.i.d. binomial market
 regime_market.py regime-switching binomial market
 dynamic_programming.py exact backward-induction CRRA solver

continuous/

black_scholes.py Merton closed form vs. duality FOC
 jump_diffusion.py Esscher EMM family, minimal entropy

constraints/

support_function.py box-constraint support function
 constrained_black_scholes.py constrained duality, auxiliary market
 dual_gradient.py independent numerical cross-check

tests/

98 unit and regression tests

Installation

```
python3 -m venv .venv
source .venv/bin/activate
pip install -e "[dev]"
```

Running the test suite

```
python -m pytest tests/ -v
```

As of the version of the code underlying this thesis, this runs 98 tests, covering every theoretical result stated as a theorem or proposition in Chapters 3 to 5 (first-order conditions, duality identities, myopia and hedging-demand properties, martingale conditions, support-function identities), together with the two regression tests documented in Chapter 7 for the bugs found during development.

Running the notebooks

```
jupyter nbconvert --to notebook --execute --inplace "Part 1/01_....ipynb"
```

or, interactively, via `jupyter notebook` or any editor with Jupyter support; every notebook is self-contained and depends only on the installed `optiport` package and standard scientific Python libraries (`numpy`, `scipy`, `matplotlib`).

Correspondence between thesis claims and tests

For traceability, Table 1 maps the main theoretical claims of the thesis to the test (or notebook) that validates them numerically.

Claim	Test / notebook
Discrete first-order condition $X_T^* = I(y dQ^*/dP)$ (Ch. 3)	Appendix B (hand); <code>test_discrete_dp.py</code>
Log utility is always myopic, i.i.d. or not (Ch. 3, Samuelson/Merton)	<code>test_discrete_dp.py</code> ; Notebook 01
Continuous-time duality = Merton HJB, complete market (Ch. 4)	<code>test_merton_validation.py</code> ; Notebook 02
Exponential utility dual optimizer = minimal entropy measure (Ch. 4)	<code>test_jump_diffusion*.py</code> ; Notebook 03
Constrained duality = unconstrained at $K = R^d$ (Ch. 5)	<code>test_constrained.py</code> (non-binding cases)
Constrained FOC via auxiliary market and $\delta(\nu^*)$ (Ch. 5)	<code>test_constrained.py</code> ; Notebook 04
Numerical algorithm agrees with closed form, no shared code (Ch. 6)	<code>test_dual_gradient.py</code> ; Notebook 05

TABLE 1 – Correspondence between theoretical claims and their numerical validation.